Evaluation

Technically, there are very few steps to run it on GPUs, elsewhere (ie. on Lamini).

```
finetuned_model = BasicModelRunner(
    "lamini/lamini_docs_finetuned"
)
finetuned_output = finetuned_model(
    test_dataset_list # batched!
)
```

Let's look again under the hood! This is the open core code of Lamini's `llama` library :)

In [2]:

```
import datasets
import tempfile
import logging
import random
import config
import os
import yaml
import logging
import difflib
import pandas as pd

import transformers
import datasets
import torch

from tqdm import tqdm
from utilities import *
from transformers import AutoTokenizer, AutoModelForCausalLM

logger = logging.getLogger(__name__)
global_config = None
```

In [3]:
```
dataset = datasets.load_dataset("lamini/lamini_docs")

test_dataset = dataset["test"]
```

```
Downloading readme:   0%|          | 0.00/577 [00:00<?, ?B/s]

Downloading data files:   0%|          | 0/2 [00:00<?, ?it/s]

Downloading data:   0%|          | 0.00/615k [00:00<?, ?B/s]

Downloading data:   0%|          | 0.00/83.7k [00:00<?, ?B/s]

Extracting data files:   0%|          | 0/2 [00:00<?, ?it/s]

Generating train split:   0%|          | 0/1260 [00:00<?, ? examples/s]

Generating test split:   0%|          | 0/140 [00:00<?, ? examples/s]
```

In [4]:
```
print(test_dataset[0]["question"])
print(test_dataset[0]["answer"])
```

```
Can Lamini generate technical documentation or user manuals for software projec
ts?
Yes, Lamini can generate technical documentation and user manuals for software
projects. It uses natural language generation techniques to create clear and co
ncise documentation that is easy to understand for both technical and non-techn
ical users. This can save developers a significant amount of time and effort in
creating documentation, allowing them to focus on other aspects of their projec
ts.
```

In [5]:
```
model_name = "lamini/lamini_docs_finetuned"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForCausalLM.from_pretrained(model_name)
```

```
/usr/local/lib/python3.9/site-packages/huggingface_hub/file_download.py:943: Fu
tureWarning: `resume_download` is deprecated and will be removed in version 1.
0.0. Downloads always resume when possible. If you want to force a new downloa
d, use `force_download=True`.
  warnings.warn(

tokenizer_config.json:   0%|          | 0.00/264 [00:00<?, ?B/s]

tokenizer.json: 0.00B [00:00, ?B/s]

special_tokens_map.json:   0%|          | 0.00/99.0 [00:00<?, ?B/s]

Special tokens have been added in the vocabulary, make sure the associated word
embeddings are fine-tuned or trained.

config.json:   0%|          | 0.00/717 [00:00<?, ?B/s]

model.safetensors:   0%|          | 0.00/282M [00:00<?, ?B/s]

generation_config.json:   0%|          | 0.00/111 [00:00<?, ?B/s]
```

Setup a really basic evaluation function

In [6]:
```
def is_exact_match(a, b):
    return a.strip() == b.strip()
```

In [7]:
```
model.eval()
```

```
GPTNeoXForCausalLM(
  (gpt_neox): GPTNeoXModel(
    (embed_in): Embedding(50304, 512)
    (emb_dropout): Dropout(p=0.0, inplace=False)
    (layers): ModuleList(
      (0-5): 6 x GPTNeoXLayer(
        (input_layernorm): LayerNorm((512,), eps=1e-05, elementwise_affine=Tru
e)
        (post_attention_layernorm): LayerNorm((512,), eps=1e-05, elementwise_af
fine=True)
        (post_attention_dropout): Dropout(p=0.0, inplace=False)
        (post_mlp_dropout): Dropout(p=0.0, inplace=False)
        (attention): GPTNeoXAttention(
          (rotary_emb): GPTNeoXRotaryEmbedding()
          (query_key_value): Linear(in_features=512, out_features=1536, bias=Tr
ue)
          (dense): Linear(in_features=512, out_features=512, bias=True)
          (attention_dropout): Dropout(p=0.0, inplace=False)
        )
        (mlp): GPTNeoXMLP(
          (dense_h_to_4h): Linear(in_features=512, out_features=2048, bias=Tru
e)
          (dense_4h_to_h): Linear(in_features=2048, out_features=512, bias=Tru
e)
          (act): GELUActivation()
        )
      )
    )
    (final_layer_norm): LayerNorm((512,), eps=1e-05, elementwise_affine=True)
  )
  (embed_out): Linear(in_features=512, out_features=50304, bias=False)
)
```

In [8]:
```
def inference(text, model, tokenizer, max_input_tokens=1000, max_output_tok
  # Tokenize
  tokenizer.pad_token = tokenizer.eos_token
  input_ids = tokenizer.encode(
      text,
      return_tensors="pt",
      truncation=True,
      max_length=max_input_tokens
  )

  # Generate
  device = model.device
  generated_tokens_with_prompt = model.generate(
    input_ids=input_ids.to(device),
    max_length=max_output_tokens
  )

  # Decode
  generated_text_with_prompt = tokenizer.batch_decode(generated_tokens_with

  # Strip the prompt
  generated_text_answer = generated_text_with_prompt[0][len(text):]

  return generated_text_answer
```

Run model and compare to expected answer

In [9]:
```
test_question = test_dataset[0]["question"]
generated_answer = inference(test_question, model, tokenizer)
print(test_question)
print(generated_answer)
```

```
The attention mask and the pad token id were not set. As a consequence, you may
observe unexpected behavior. Please pass your input's `attention_mask` to obtai
n reliable results.
Setting `pad_token_id` to `eos_token_id`:0 for open-end generation.

Can Lamini generate technical documentation or user manuals for software projec
ts?
Yes, Lamini can generate technical documentation or user manuals for software p
rojects. This can be achieved by providing a prompt for a specific technical qu
estion or question to the LLM Engine, or by providing a prompt for a specific t
echnical question or question. Additionally, Lamini can be trained on specific
technical questions or questions to help users understand the process and provi
de feedback to the LLM Engine. Additionally, Lamini
```

In [10]:
```
answer = test_dataset[0]["answer"]
print(answer)
```

```
Yes, Lamini can generate technical documentation and user manuals for software
projects. It uses natural language generation techniques to create clear and co
ncise documentation that is easy to understand for both technical and non-techn
ical users. This can save developers a significant amount of time and effort in
creating documentation, allowing them to focus on other aspects of their projec
ts.
```

In [11]:
```
exact_match = is_exact_match(generated_answer, answer)
print(exact_match)
```

```
False
```

Run over entire dataset

In [12]:

```
n = 10
metrics = {'exact_matches': []}
predictions = []
for i, item in tqdm(enumerate(test_dataset)):
    print("i Evaluating: " + str(item))
    question = item['question']
    answer = item['answer']

    try:
      predicted_answer = inference(question, model, tokenizer)
    except:
      continue
    predictions.append([predicted_answer, answer])

    #fixed: exact_match = is_exact_match(generated_answer, answer)
    exact_match = is_exact_match(predicted_answer, answer)
    metrics['exact_matches'].append(exact_match)

    if i > n and n != -1:
      break
print('Number of exact matches: ', sum(metrics['exact_matches']))
```

```
0it [00:00, ?it/s]The attention mask and the pad token id were not set. As a
consequence, you may observe unexpected behavior. Please pass your input's `a
ttention_mask` to obtain reliable results.
Setting `pad_token_id` to `eos_token_id`:0 for open-end generation.

i Evaluating: {'question': 'Can Lamini generate technical documentation or us
er manuals for software projects?', 'answer': 'Yes, Lamini can generate techn
ical documentation and user manuals for software projects. It uses natural la
nguage generation techniques to create clear and concise documentation that i
s easy to understand for both technical and non-technical users. This can sav
e developers a significant amount of time and effort in creating documentatio
n, allowing them to focus on other aspects of their projects.', 'input_ids':
[5804, 418, 4988, 74, 6635, 7681, 10097, 390, 2608, 11595, 84, 323, 3694, 649
3, 32, 4374, 13, 418, 4988, 74, 476, 6635, 7681, 10097, 285, 2608, 11595, 84,
323, 3694, 6493, 15, 733, 4648, 3626, 3448, 5978, 5609, 281, 2794, 2590, 285,
44003, 10097, 326, 310, 3477, 281, 2096, 323, 1097, 7681, 285, 1327, 14, 4874
6, 4212, 15, 831, 476, 5321, 12259, 247, 1534, 2408, 273, 673, 285, 3434, 27
5, 6153, 10097, 13, 6941, 731, 281, 2770, 327, 643, 7794, 273, 616, 6493, 1
5], 'attention_mask': [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
```

In [13]:
```
df = pd.DataFrame(predictions, columns=["predicted_answer", "target_answer"
print(df)
```

```
                                      predicted_answer  \
0   Yes, Lamini can generate technical documentati...
1   You can use the Authorization HTTP header to g...
2   Lamini’s LLM Engine is a LLM Engine for develo...
3   Yes, there is a community or support forum ava...
4   Yes, the Lamini library can be utilized for te...
5   Lamini is designed to be flexible and scalable...
6   You can instantiate the LLM engine using the L...
7   Yes, Lamini provides mechanisms for compressio...
8   The performance of LLMs trained using Lamini c...
9   Yes, there is a support and community availabl...
10  Yes, there are some code samples available in ...
11  Yes, there are some code samples available tha...

                                        target_answer
0   Yes, Lamini can generate technical documentati...
1   The Authorization HTTP header should include t...
2   Yes, the code includes a version parameter in ...
3   Yes, there is a community forum available for ...
4   The Lamini library is not specifically designe...
5   Lamini offers both free and paid plans for usi...
6   You can instantiate the LLM engine using the l...
7   Yes, Lamini provides mechanisms for model comp...
8   According to the information provided, Lamini ...
9   Yes, there is a support community available to...
10  Yes, the Python logging module documentation p...
11  Yes, there is a separate section in the docume...
```

Evaluate all the data

In [14]:
```
evaluation_dataset_path = "lamini/lamini_docs_evaluation"
evaluation_dataset = datasets.load_dataset(evaluation_dataset_path)
```

```
Downloading readme:   0%|          | 0.00/413 [00:00<?, ?B/s]

Downloading data files:   0%|          | 0/1 [00:00<?, ?it/s]

Downloading data:   0%|          | 0.00/86.1k [00:00<?, ?B/s]

Extracting data files:   0%|          | 0/1 [00:00<?, ?it/s]

Generating train split:   0%|          | 0/139 [00:00<?, ? examples/s]
```

In [15]:
```
pd.DataFrame(evaluation_dataset)
```

	train
0	{'predicted_answer': 'Yes, Lamini can generate...
1	{'predicted_answer': 'You can use the Authoriz...
2	{'predicted_answer': 'Lamini's LLM Engine is a...
3	{'predicted_answer': 'Yes, there is a communit...
4	{'predicted_answer': 'Yes, the Lamini library ...
...	...
134	{'predicted_answer': 'Yes, Lamini has the abil...
135	{'predicted_answer': 'I wish! This documentati...
136	{'predicted_answer': 'Yes, Lamini can generate...
137	{'predicted_answer': 'Yes, the documentation h...
138	{'predicted_answer': 'Yes, you can learn to us...

139 rows × 1 columns

Try the ARC benchmark

This can take several minutes

In [16]:
```
!python lm-evaluation-harness/main.py --model hf-causal --model_args pretra
```

```
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true
| false)
```

```
Selected Tasks: ['arc_easy']
Using device 'cpu'
/usr/local/lib/python3.9/site-packages/huggingface_hub/file_download.py:943: Fu
tureWarning: `resume_download` is deprecated and will be removed in version 1.
0.0. Downloads always resume when possible. If you want to force a new downloa
d, use `force_download=True`.
  warnings.warn(
Special tokens have been added in the vocabulary, make sure the associated word
embeddings are fine-tuned or trained.
Downloading readme: 9.00kB [00:00, 8.46MB/s]
Downloading data files:   0%|                                | 0/3 [00:00<?, ?it/
s]
Downloading data:   0%|                                  | 0.00/331k [00:00<?, ?B/
s]
Downloading data: 100%|████████████████████| 331k/331k [00:00<00:00, 605kB/
s]
Downloading data files:  33%|██████              | 1/3 [00:00<00:01,  1.82it/
s]
Downloading data:   0%|                                  | 0.00/346k [00:00<?, ?B/
s]
Downloading data: 100%|████████████████████| 346k/346k [00:00<00:00, 693kB/
s]
Downloading data files:  67%|████████████       | 2/3 [00:01<00:00,  1.92it/
s]
Downloading data:   0%|                                  | 0.00/86.1k [00:00<?, ?B/
s]
Downloading data: 100%|████████████████████| 86.1k/86.1k [00:00<00:00, 439kB/
s]
Downloading data files: 100%|████████████████████| 3/3 [00:01<00:00,  2.40it/
s]
Extracting data files: 100%|████████████████████| 3/3 [00:00<00:00, 3005.95it/
s]
Generating train split: 100%|████| 2251/2251 [00:00<00:00, 428588.60 examples/
s]
Generating test split: 100%|█████| 2376/2376 [00:00<00:00, 672628.67 examples/
s]
Generating validation split: 100%|█| 570/570 [00:00<00:00, 344836.76 examples/
s]
Task: arc_easy; number of docs: 2376
Task: arc_easy; document 0; context prompt (starting on next line):
Question: Which is the function of the gallbladder?
Answer:
(end of prompt on previous line)
Requests: [Req_loglikelihood('Question: Which is the function of the gallbladde
r?\nAnswer:', ' store bile')[0]
, Req_loglikelihood('Question: Which is the function of the gallbladder?\nAnswe
r:', ' produce bile')[0]
, Req_loglikelihood('Question: Which is the function of the gallbladder?\nAnswe
r:', ' store digestive enzymes')[0]
, Req_loglikelihood('Question: Which is the function of the gallbladder?\nAnswe
r:', ' produce digestive enzymes')[0]
]
Running loglikelihood requests
100%|████████████████████████████████████| 9496/9496 [03:35<00:00, 44.12it/
s]
{
  "results": {
```

```
    "arc_easy": {
      "acc": 0.3122895622895623,
      "acc_stderr": 0.009509325983631455,
      "acc_norm": 0.3135521885521885,
      "acc_norm_stderr": 0.009519779157242258
    }
  },
  "versions": {
    "arc_easy": 0
  },
  "config": {
    "model": "hf-causal",
    "model_args": "pretrained=lamini/lamini_docs_finetuned",
    "num_fewshot": 0,
    "batch_size": null,
    "batch_sizes": [],
    "device": "cpu",
    "no_cache": false,
    "limit": null,
    "bootstrap_iters": 100000,
    "description_dict": {}
  }
}
hf-causal (pretrained=lamini/lamini_docs_finetuned), limit: None, provide_descr
iption: False, num_fewshot: 0, batch_size: None
|  Task  |Version| Metric |Value |   |Stderr|
|--------|------:|--------|-----:|---|-----:|
|arc_easy|      0|acc     |0.3123|±  |0.0095|
|        |       |acc_norm|0.3136|±  |0.0095|
```